a lot of value that can be uncovered using this technology," Pai adds.

Immense opportunities

Drone technology is disrupting the way mining operations, data collection, and insight generation in mining are transformed. "I believe any technology that simplifies the operation of a mine, and makes the life of a geologist mining engineer safe, is here to stay to prosper, and drones are one such technology," says Ajit Sahu, Chief of Exploration and Technology, Vedanta Ltd.

That being said, there is a long way the technology needs to travel, especially in terms of its impact on greenfield exploration. "While the easier ones have been discovered, the less easy ones are yet to be made. I would like to see some breakthrough happening in greenfield exploration use of drones, where the inaccessible become accessible for a geologist and the difficult terrains are traversed," adds Sahu.

It is, therefore, crucial to understand where the mining process lacks behind and what the drone industry can particularly help them tackle.

Common issues faced in mines

Here are some areas where the drones can help:

Blasting. Measuring bench geometry before and after the blasting process is an essential part of any mining exercise. The traditional method of using total stations is slow, cumbersome and puts personnel in mortal danger. (A total station is used to record the absolute location of the tunnel walls, ceilings (backs), and floors as the drifts of an underground mine are driven.) There is no way to verify if drill holes for explosives are positioned as planned. Drones can help by reducing cost and time overruns due to improper management of men, material, and safety surveillance.

Output. Miners have historically made rough assumptions as to the volume of outputs generated, which leads to underreported numbers. Any mistake at the bottom of the funnel leads to resource mismanagement at a macro operational level. A drone can measure critical volumes 2-3 times faster from source to stockpile in near real-time. It can monitor volumes of pits, overburden, dumps, stockpiles, and other critical volumes across the mine 2-3 times faster.

Transport. The design of haul roads and systems using satellite imagery is inefficient as the data is outdated and inaccurate to the extent of several metres. Improperly designed transport routes lead to loss in output production and operating revenues. Drones being highly-accurate, can reduce automated post-processing errors to the minimum. Stockpile volumes can be estimated with an accuracy of 99%.

The challenge of integrating software and hardware

The introduction of drones in mining industry has yielded path-breaking results even in the short amount of time that they have been around. However, as is the case with any nascent technology, there are certain voids that drones are yet to fill. "To my mind, it's not the drones which is the most important thing but how we use the software," remarks Piyush Srivastava, Chief of Natural Resources at Tata Steel.

Drones so far have only aided in the collection and capture of data. Even though drones are capable of carrying both the laser cameras as well as multispectral cameras, operators are yet to learn the integration of software with the data set that they receive from drones to turn them into usable data.

One of the biggest challenges that photogrammetry cloud data collection poses is the trouble of dealing with its sheer size. As per Srivastava, the size of the data for scanning up to five square kilometres of land can reach up to 60GB. Software that the industry currently uses widely is yet to catch up with the demands of such heavy data. "Now to process the data for when there are images from multiple file paths and then you do cognition and then aero-triangulation, the speed is a limiting factor. Sometimes it takes a few days to process the data," notes Srivastava.

One of the possible solutions to deal with this could be the adoption of a hub-and-spoke model, where

Drones aiding in magnetic surveys

The process of mineral discovery and block development in mining is a long and winding process. Surveying is the first phase of the discovery process. This includes geological mapping, physical mapping, and geochemical mapping. All these datasets narrow down the area from there, so that a 10,000-15,000sq km of metallic deposit site can be condensed to 100sq km. As this process continues, the deposit area is further narrowed down to a 1x1sq km of mineral deposit.

This, as per Dr D.S. Jeere, Director of Geology at GSI, is where drones can come in handy. He says, "I find that there is a very big role for drones to play here because what we need in this 1sq km is a 1x4000 scale data graphic mapping."

A high-resolution magnetic survey is another area where drones, especially with their small sensors, can play a major role in the mine development process. But the quality of sensors plays a major role in how the end data comes out.

Sahu says, "We did a drone-aided magnetic survey recently and found that it is good in terms of well-located data. It could be done in a small time period; whatever objectives were set were met. But I also know that the quality of data was not great. It could be because of the sensor. In terms of the survey, one of the problems that we had in our mind was that we may struggle to find well-located data. But that was not the case—it turned out to be good."